

How Technology Paints Buildings Green

Most green buildings today work with independently controlled and managed lighting, HVAC and other loads. Some very advanced buildings provide loose integration at the data visualisation or dashboard level. This allows continuous monitoring of the different loads and corrective actions as and when desired

UMA GUPTA

Construction and functioning of buildings has a profound effect on the environment. Buildings account for 40 per cent

of our total energy requirements. These need to save energy, in order to reduce CO₂ output and protect the environment.

In view of this, constructing green buildings is becoming a standard practice today. Compared to conventional

buildings, a green building promotes healthier living spaces for the occupants while ensuring that utility wastage (water, energy and other natural resources) is minimal.

A smart green building can be described as the convergence of technology and real estate applications. Devices and solutions are transferred onto a converged IP network disrupting traditional design, transforming the physical space into a service offering.

Integrated energy management systems support sophisticated power management and energy conservation, leading to long-term environmental sustainability. Lighting automation solutions are delivered over the converged IP network. Centralised monitoring and control of elevators, heating, ventilation and cooling (HVAC), and facilities help ensure that the building meets or even exceeds performance standards. An added benefit is that with the knowledge of energy usage patterns, building operators can respond to power cuts and proactively manage the carbon footprint to build a sustainable and more energy-efficient environment.

Converging multiple functions onto system devices means reducing the number of IT equipment needed, use of systems that consume less power, and more efficient use of all equipment. This, in turn, makes a significant difference in operating costs.

What goes inside

The most advanced green buildings deploy a host of IT and electronic ele-

